

Windows Processes — THM Compact Cheat Sheet

Tasks 2-12 • Quick reference • Process tree • Detection notes

PROCESS MAP

System — PID 4	smss.exe
csrss.exe	wininit.exe
services.exe	svchost.exe
lsass.exe	winlogon.exe
userinit.exe	explorer.exe

Key chains: wininit → services → svchost | smss → csrss / winlogon | winlogon → userinit → explorer

PROCESS QUICK SHEET

Process	Purpose	Normal / Key Notes
System	Hosts system threads	PID 4 • Kernel Mode only • Ntoskrnl/drivers
smss.exe	Session Manager	Early boot • creates sessions • launches early processes
csrss.exe	Client Server Runtime	User Mode • Win32 subsystem • console • process/thread lifecycle • critical
wininit.exe	Windows Initialization	Session 0 • launches services.exe, lsass.exe, lsaiso.exe • usually 1
services.exe	Service Control Manager	Manages Windows Services • child of wininit
svchost.exe	Service Host	Hosts services/DLLs • many instances normal • -k = service group
lsass.exe	Local Security Authority	Authentication/credentials/security • SYSTEM • critical
winlogon.exe	Windows Logon	SAS (Ctrl+Alt+Del) • profile • lock • SYSTEM • System32 • one+ sessions
userinit.exe	User initialization	Loads user shell
explorer.exe	Windows shell	Desktop • taskbar • File Explorer • normal shell

SYSTEM / KERNEL NOTES

PID 4	System Thread	Kernel Mode
No user address space	Paged Pool	Nonpaged Pool

System thread: kernel-mode-only thread executing kernel/driver code. Paged Pool: kernel memory that may be paged. Nonpaged Pool: kernel memory that must remain resident.

SESSION + LOGON

Session 0 → Services/System	Session 1+ → User sessions	smss.exe → session setup
winlogon.exe → logon	NTUSER.DAT → HKCU	userinit.exe → explorer.exe

SAS: Secure Attention Sequence = Ctrl + Alt + Delete. winlogon: loads user profile, handles locking/screensaver, and participates in logon.

SVCHOST — HIGH VALUE NOTES

svchost.exe	-k Group	ServiceDLL
Registry	Many instances = normal	Masquerading risk

ServiceDLL: HKLM\SYSTEM\CurrentControlSet\Services\<Service>\Parameters\ServiceDLL

Check: image path → command line → -k group → ServiceDLL → DLL path. Attackers may use svchost.exe, misspell it (scvhost.exe), or install a malicious service DLL.

NORMAL VS SUSPICIOUS

Process	Verify	Red Flags
System	PID 4 / kernel context	Unexpected user-mode behavior
smss.exe	Early boot / expected path	Unexpected location/behavior
csrss.exe	System32 / critical	Fake path/name
wininit.exe	Usually one / Session 0	Unexpected child/path
services.exe	Parent = wininit	Wrong parent/path

svchost.exe	System32 / -k / ServiceDLL	Fake path/name / odd service DLL
lsass.exe	SYSTEM / System32	Wrong user/path / suspicious access
winlogon.exe	SYSTEM / System32 / sessions	Actual unexpected parent / wrong path / wrong shell
explorer.exe	User session / shell	Unexpected path / suspicious shell persistence

SMART TERMS

Term	Definition
Process	Running program with its own resources/address space.
Thread	Execution unit inside a process.
Instance	Separate running process/copy of the same executable.
Parent / Child	Creator process / process it creates.
Session	Windows execution/user context.
User Mode	Restricted application execution environment.
Kernel Mode	Privileged kernel/driver execution environment.
Handle	OS-managed reference to an object/resource.
Masquerading	Malware pretending to be a legitimate process.
ServiceDLL	Registry value pointing to a service implementation DLL.

INVESTIGATION CHECKLIST

1. Image Name	2. Image Path	3. Parent / Child	4. User / Account
5. Command Line	6. Start Time	7. Registry / Service	8. Behavior

Rule: Never judge a process by its name alone. Correlate path + parent/child + account + command line + configuration + behavior.